INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Google NBY-2, OH -- station KCMH, America/New_York
Committed pair	OSM 1252196811 -> 1222007663 Google NBY-3 / Google NBY-2
Facade gap	81.3 m between the two halls
Weather record	43,786 real hourly records from KCMH
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.4779 C at 235 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2024-02-28 (crossing)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 1 mode change. That is 3 more than the reactive incumbent, at 0 unsafe hours against its 2. 5 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	8 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	5 of 24 hours free cooling, 2 mode change(s), 2 unsafe hour(s)
	5 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dew point	15.837	18.0	16.415	4.106
01	mechanical	no	dew point	15.846	18.0	16.415	3.693
02	mechanical	no	dew point	16.440	18.0	16.975	3.731
03	mechanical	yes	switch budget	16.872	18.0	18.645	3.351
04	mechanical	yes	switch budget	16.440	18.0	19.232	2.958
05	mechanical	yes	switch budget	16.719	18.0	17.589	2.614
06	mechanical	yes	switch budget	17.472	18.0	15.207	2.745
07	mechanical	yes	switch budget	17.557	18.0	15.373	2.365
08	mechanical	no	dry-bulb	20.660	18.0	14.907	2.938
09	mechanical	no	dry-bulb	25.082	18.0	10.560	3.903
10	mechanical	no	dry-bulb	28.090	18.0	6.671	5.024
11	mechanical	no	dry-bulb	28.647	18.0	4.441	5.880

12	mechanical	no	dry-bulb	27.462	18.0	3.330	5.981
13	mechanical	no	dry-bulb	27.840	18.0	2.783	5.851
14	mechanical	no	dry-bulb	25.541	18.0	1.673	4.950
15	mechanical	no	dry-bulb	19.944	18.0	1.673	4.541
16	FREE	yes	-	13.046	18.0	0.560	3.358
17	FREE	yes	-	9.265	18.0	0.001	3.422
18	FREE	yes	-	5.817	18.0	-1.110	2.751
19	FREE	yes	-	4.204	18.0	-1.670	3.194
20	FREE	yes	-	1.889	18.0	-2.220	3.399
21	FREE	yes	-	1.379	18.0	-2.780	3.796
22	FREE	yes	-	0.321	18.0	-3.890	4.293
23	FREE	yes	-	-0.150	18.0	-3.771	4.359

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.837 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

01:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 15.846 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

02:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 16.440 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.872 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.440 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.719 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.472 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.557 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.660 C against a 18.0 C limit, so it fails by 2.660 C. It would take a limit 2.660 C higher, or a bound 2.660 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.082 C against a 18.0 C limit, so it fails by 7.082 C. It would take a limit 7.082 C higher, or a bound 7.082 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.090 C against a 18.0 C limit, so it fails by 10.090 C. It would take a limit 10.090 C higher, or a bound 10.090 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.647 C against a 18.0 C limit, so it fails by 10.647 C. It would take a limit 10.647 C higher, or a bound 10.647 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.463 C against a 18.0 C limit, so it fails by 9.463 C. It would take a limit 9.463 C higher, or a bound 9.463 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.840 C against a 18.0 C limit, so it fails by 9.840 C. It would take a limit 9.840 C higher, or a bound 9.840 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.541 C against a 18.0 C limit, so it fails by 7.541 C. It would take a limit 7.541 C higher, or a bound 7.541 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.943 C against a 18.0 C limit, so it fails by 1.943 C. It would take a limit 1.943 C higher, or a bound 1.943 C tighter, to change this hour.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.046 C, which is 4.954 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.358 C, and the margin is measured, not chosen: 3.102 C of group-conditional forecast error for this hour of day, plus 0.2558 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.265 C, which is 8.735 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.422 C, and the margin is measured, not chosen: 3.096 C of group-conditional forecast error for this hour of day, plus 0.3259 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 5.817 C, which is 12.183 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.751 C, and the margin is measured, not chosen: 2.494 C of group-conditional forecast error for this hour of day, plus 0.2571 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.204 C, which is 13.796 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.194 C, and the margin is measured, not chosen: 2.880 C of group-conditional forecast error for this hour of day, plus 0.3141 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.889 C, which is 16.111 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.399 C, and the margin is measured, not chosen: 3.181 C of group-conditional forecast error for this hour of day, plus 0.2185 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.379 C, which is 16.621 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.796 C, and the margin is measured, not chosen: 3.527 C of group-conditional forecast error for this hour of day, plus 0.2690 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.321 C, which is 17.679 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.293 C, and the margin is measured, not chosen: 3.973 C of group-conditional forecast error for this hour of day, plus 0.3207 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.150 C, which is 18.150 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.359 C, and the margin is measured, not chosen: 4.027 C of group-conditional forecast error for this hour of day, plus 0.3317 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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